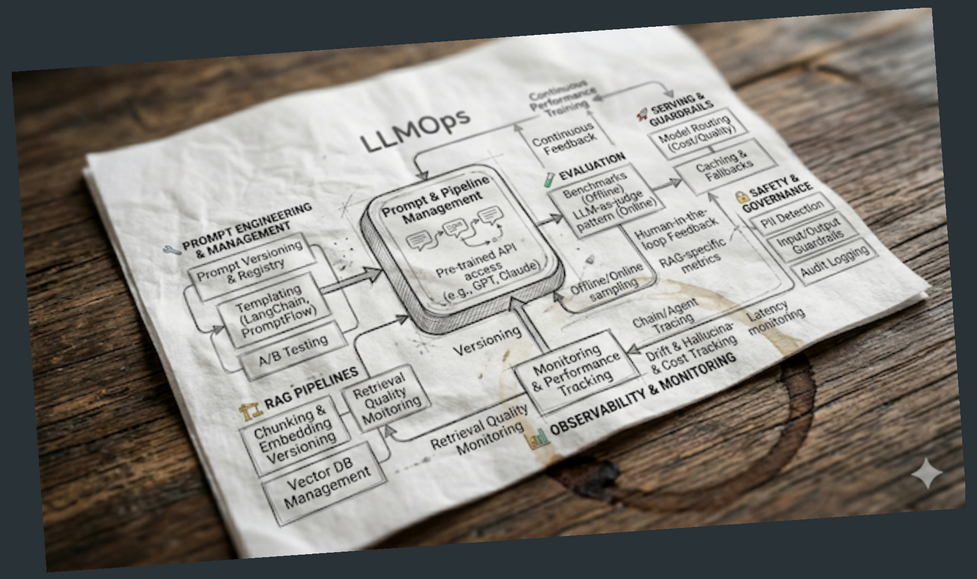


kokchun giang

LLMOps - managing the lifecycle of LLM development

a tale of a **customer support bot**

billing

account management

technical troubleshooting

feature explanation

RAG pipeline → based on GPT-5

a great **product launch** with happy team and happy customers

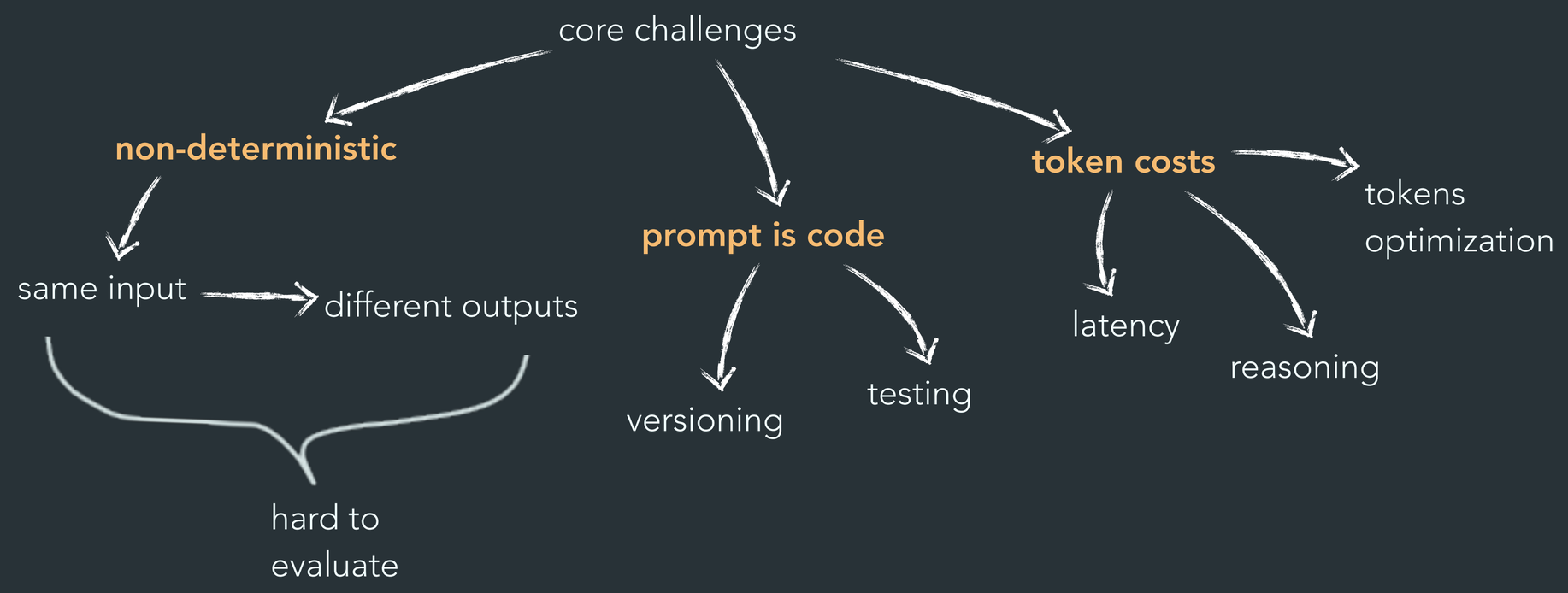
month 2
30 new features shipped
new price plan
no update knowledge base
bot answer with old knowledge
month 3
prompt change
make responses more concise
bot starts truncate troubleshooting steps
notice after 4 days due to support ticket spike
month 4
costs spiral
no rate limiting
some users asks too much
no tracking of token usage
month 5
team wants to improve bot
hard because of
no prompt versioning
no feedback signal
no structured logging
no token tracking

traditional software fails **loudly**

LLMs fail **silently** and degrades over time

LLMOps closes the gap between "system is up" and "system doing what you think it is doing"

some core challenges of **LLMOps**



key concepts of **LLMOps**

prompt engineering
- ✓ versioning
- ✓ prompt templating

observability
- ✓ tracing
- ✓ detect drift, hallucinations, toxicity
- ✓ latency monitoring

deployment
- ✓ model routing
- ✓ fallback chans
- ✓ semantic caching
- ✓ guardrails

evaluation
- ✓ LLM as judge
- ✓ human-in-the-loop
- ✓ offline evals
- ✓ online evals

RAG pipelines
- ✓ chunking strategies
- ✓ vector db management
- ✓ embedding model versioning
- ✓ monitor retrieval quality

safety & governance
- ✓ PII detection
- ✓ audit logging
- ✓ rate limiting